



ZHEJIANG JINFANDA BIOCHEMICAL CO.,LTD

Address: Jinfanda Tower, 98 Tongjun Lu, Tonglu County, Zhejiang Province, China

Tell: 0571- 56986662 87245868 Fax: 0571-87245818 56986618

Material Safety Data Sheet

1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND THE COMPANY/UNDERTAKING

Product information

Product name	Glyphosate Ammonium Salt 88.8% SG
Chemical class	Glycine derivative
Usage	Non-selective systemic herbicide
Manufacturer	Zhejiang Jinfanda Biochemical Co.,Ltd
Telephone	0571-56986662 87245868
Telefax	0571-56986618 87245818

2. COMPOSITION / INFORMATION ON INGREDIENTS

Components

Component	CAS-No	Concentration
Glyphosate Ammonium	40465-66-5	88.8%
Inert Ingredients	/	11.2%

Components not precisely identified are proprietary or not hazardous.

3. HAZARDS IDENTIFICATION

Emergency Overview

Warning statement:

Keep put of the reach of children.

Warning!

Causes eye irritation. Harmful if swallowed or inhaled. May cause skin irritation.

Do not get in eyes or clothing. Avoid breathing vapor or spray mist.

Potential Health Effects

Inhalation: No occupational exposure to this product has been reported to cause respiratory tract irritation.

Eye Irritation: Direct contact may cause temporary eye irritation and conjunctivitis.

Skin Absorption: Prolonged contact may cause skin irritation. Hazards Identification continued:

Ingestion: Ingestion has been reported to produce gastro-intestinal discomfort with irritation of the mouth, nausea, vomiting, and diarrhea. Oral ingestion of large



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quantities of concentrated product has been reported to result in hypotension and pulmonary edema.

4. FIRST AID MEASURES

Inhalation

If inhaled, remove the person to fresh air and keep warm and at rest. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Skin contact

Remove contaminated clothing, shoes and leather goods. Wash skin gently and thoroughly with plenty of water or shower. Seek medical attention if necessary.

Eye contact

Hold eye open and Immediately flush the eyes with gently flowing lukewarm water or saline solution for 15-20 minutes until no evidence of chemical remains. Remove contact lenses, if present, after the first 5 minutes, then continue flushing. Call a poison control center or doctor for treatment advice.

Ingestion

This material will produce gastrointestinal irritation. Immediately dilute by swallowing water or milk. Refer for medical attention. Do not give anything by mouth to an unconscious or semi conscious person.

Notes to physician

This product is not an inhibitor of cholinesterase, treatment with atropine and oximes is not indicated.

No specific antidote is available. If poisoning is suspected apply symptomatic therapy. Contact Poisons Information Center for more information.

5. FIRE FIGHTING MEASURES

Flash point :

Not flammable

Extinguishing media:

Recommended: Water, foam, dry chemical, carbon dioxide (CO₂)

Fire fighting instructions:

Firefighter or other who may be exposed to vapors or products of combustion should wear full protective clothing and self-contained breathing apparatus.

Equipment should be thoroughly decontaminated after each use.

Unusual fire and explosion hazards:

Minimise use of water to prevent environmental contamination.

Hazardous products of combustion

Carbon monoxide (CO), phosphorus oxides (P_xO_y), nitrogen oxides (NO_x).

6. ACCIDENTAL RELEASE MEASURES



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Environmental precautions

SMALL QUANTITIES:

Low environmental hazard.

LARGE QUANTITIES:

Minimise spread.

Keep out of drains, sewers, ditches and water ways.

Notify authorities.

Spill or Leak Procedures:

Liquid spills on the floor or other impervious surfaces should be contained or diked, and should be absorbed with attapulgite, bentonite, or other absorbent clays.

Spill or leak procedures:

Collect contaminated absorbent, place in plastic-lined metal drum, and dispose of in accordance with instructions provided under DISPOSAL. Thoroughly scrub floor with a strong industrial type detergent solution and rinse with warm water. Liquid spills that soak into the ground should be dug-up, placed in plastic-lined metal drums, and disposed of in accordance with instructions provided under DISPOSAL. Leaking containers should be separated from non-leakers and either the container or its contents transferred to a plastic-lined drum or to other non-leaking container. Dispose of leaking container in accordance with instructions provided under disposal. Any received spilled liquid should be similarly collected and disposed of.

7. HANDLING AND STORAGE

Handling

Good industrial practice in housekeeping and personal hygiene should be followed. When using do not eat, drink or smoke. Wash hands thoroughly after handling or contact. Thoroughly clean equipment after use. Do not contaminate drains, sewers and water ways when disposing of equipment rinse water. Emptied containers retain vapour and product residue.

Storage

Minimum storage temperature: -15 °C, Maximum storage temperature: 50 °C. Compatible materials for storage: stainless steel, aluminium, fibreglass, plastic, glass lining. Incompatible materials for storage: galvanised steel, unlined mild steel, see section 10. Keep out of reach of children. Keep away from food, drink and animal feed. Keep only in the original container. Partial crystallization may occur on prolonged storage below the minimum storage temperature. If frozen, place in warm room and shake frequently to put back into solution. Minimum shelf life: 5 years. Not compatible with strong bases (such as sodium hydroxide and potassium hydroxide). Store in tightly closed containers in a cool, well-ventilated area.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

No special requirement when used as recommended.



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Personal protective equipment

Eyes: Wear chemical splash goggles and face shield.

Skin: Wear appropriate protective gloves to prevent skin exposure.

Clothing: Wear appropriate protective clothing to prevent skin exposure.

Respirators: Use an approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene measures

When using, do not eat, drink or smoke. Wash hands and face before breaks and immediately after handling the product. Handle in accordance with good industrial hygiene and safety practice.

Protective measures

Avoid contact with skin and eyes. Avoid breath vapor of product. Do not swallow. Wash hands with soap and water and rinse after handling product. Shower after each shift. Wash all work clothing and completely clean all PPE (Personal Protection Equipment) after each shift.

9. PHYSICAL AND CHEMICAL PROPERTIES

Form: Granular

Content of glyphosate ammonium (%) $\geq 88.8\%$

Color: Hermosa pink granule

Odor: Odorless

Content of formaldehyde: $\leq 0.6\text{g/kg}$

PH range: 3.0~8.0

Water content: $\leq 2.5\%$

Dust: $\leq 15\text{mg}$

Persistent foam (after 1 minute) $\leq 60\text{ml}$

Storage stability: Stable under normal recommended condition

10. STABILITY AND REACTIVITY

Incompatible Materials:

Caustic materials

Hazardous Decomposition:

Carbon oxides (COx), phosphorus oxides (PxOy), nitrogen oxides (NOx).

Hazardous Polymerization:

Material is not known to polymerize.

11. TOXICOLOGICAL INFORMATION



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Acute oral toxicity

Rat, LD₅₀: >5000 mg/kg body weight Practically non-toxic.

Acute dermal toxicity

Rat, LD₅₀ (limit test): > 2000 mg/kg body weight Practically non-toxic. No mortality.

Acute inhalation toxicity

Rat, LC₅₀, 4 hours, aerosol: >2 mg/L Practically non-toxic.

Skin irritation

Non irritation.

Eye irritation

Moderate irritation.

Skin sensitization

Weak skin sensitization.

The followed information are based on glyphosate acid active ingredient.

Mutagenicity

In vitro and in vivo mutagenicity test(s): Not mutagenic.

Repeated dose toxicity

Rabbit, dermal, 21 days: NOAEL toxicity: > 5,000 mg/kg body weight/day, Target organs/systems: none, Other effects: none

Rat, oral, 3 months:

NOAEL toxicity: > 20,000 mg/kg diet; Target organs/systems: none ; Other effects: none

Carcinogenicity

Mouse, oral, 24 months: NOEL tumour: > 30,000 mg/kg diet NOAEL toxicity: ~ 5,000 mg/kg diet Tumours: none, Target organs/systems: liver Other effects: decrease of body weight gain, histopathologic effects

Rat, oral, 24 months:

NOEL tumour: > 20,000 mg/kg diet NOAEL toxicity: ~ 8,000 mg/kg diet Tumours: none Target organs/systems: eyes Other effects: decrease of body weight gain, histopathologic effects

12. ECOLOGICAL INFORMATION

The followed information are based on glyphosate acid active ingredient.

Aquatic toxicity, fish

Rainbow trout (*Oncorhynchus mykiss*): Acute toxicity, 96 hours, static, LC₅₀: 5.4 mg/L Moderately toxic.



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Bluegill sunfish (*Lepomis macrochirus*): Acute toxicity, 96 hours, static, LC_{50} : 7.3 mg/L Moderately toxic.

Aquatic toxicity, invertebrates

Water flea (*Daphnia magna*): Acute toxicity, 48 hours, static, EC_{50} : 11 mg/L

Slightly toxic.

Avian toxicity

Mallard duck (*Anas platyrhynchos*): Dietary toxicity, 5 days, LC_{50} : > 5,620 mg/kg diet Practically non-toxic.

Bobwhite quail (*Colinus virginianus*): Dietary toxicity, 5 days, LC_{50} : > 5,620 mg/kg diet Practically non-toxic.

Arthropod toxicity

Honey bee (*Apis mellifera*): Oral/contact, 48 hours, LD_{50} : > 100 μ g/bee

Practically non-toxic.

Soil organism toxicity, invertebrates

Earthworm (*Eisenia foetida*): Acute toxicity, 14 days, LC_{50} : > 1,250 mg/kg soil

Practically non-toxic.

Dissipation

Soil, field: Half life: 2 - 174 days K_{oc} : 884 - 60,000 L/kg Adsorbs strongly to soil.

Water, aerobic: Half life: < 7 days

13. CONSIDERATION ABOUT THE PRODUCT DISPOSABILITY

Wastes resulting from the use of this product that cannot be used or chemically reprocessed should be disposed of in a landfill approved for pesticide disposal in accordance with any applicable national or local procedures. An empty container retains vapor and product residue. Observe all labeled safeguards until container is cleaned, reconditioned, or destroyed.

Metal drums:

Triple rinse container, and then offer for recycling or reconditioning, or puncture and dispose of in a sanitary landfill, or by other procedures approved by state and local authorities.

Plastic Jugs:

Do not reuse container. Triple rinse container, then puncture and dispose of in a sanitary landfill, or by incineration, or, if allowed by state and local authorities, by burning. If burned, stay out of smoke.

Metal Bulk:

Triple rinse emptied bulk containers, and then offer for recycling or reconditioning or disposal in a manner approved by state and local authorities.



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Plastic Drums/ Mini-Bulk:

Do not reuse container. Triple rinse the container, then puncture and dispose of in a sanitary landfill, or by incineration, or, if allowed by state and local authorities, by burning. If burned, stay out of smoke.

14. TRANSPORT INFORMATION

Department of Transportation (DOT):

Not Regulated

International Air Transport Association (IATA):

Not Regulated

International Maritime Organization (IMO):

Not Regulated

15. REGULATORY INFORMATION

The followed regulations on safety of dangerous chemicals had regulated the dangerous chemicals in china.

GB/T 15098-94: The principle of classification of transport packaging groups of dangerous goods

GB 13690-92: Classification and Mark of Common Dangerous Chemicals

GB 16483-2000: General rules for preparation of chemical safety data sheet (CSDS)

GB 6944-2005: Classification and Name Numeration of Dangerous Goods

GB 12268-2005: List of Dangerous chemicals

High-toxic chemicals List (2002 version)

This product had not been classified as a hazardous chemical

16. OTHER INFORMATION

Please contact us through mail to the followed mailbox for any other questions:

jianglei@jinfanda.com

DISCLAIMER

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the stability of the information for their particular purposes. In no way shall the company be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, however arising, even if the company has been advised of the possibility of such damages.